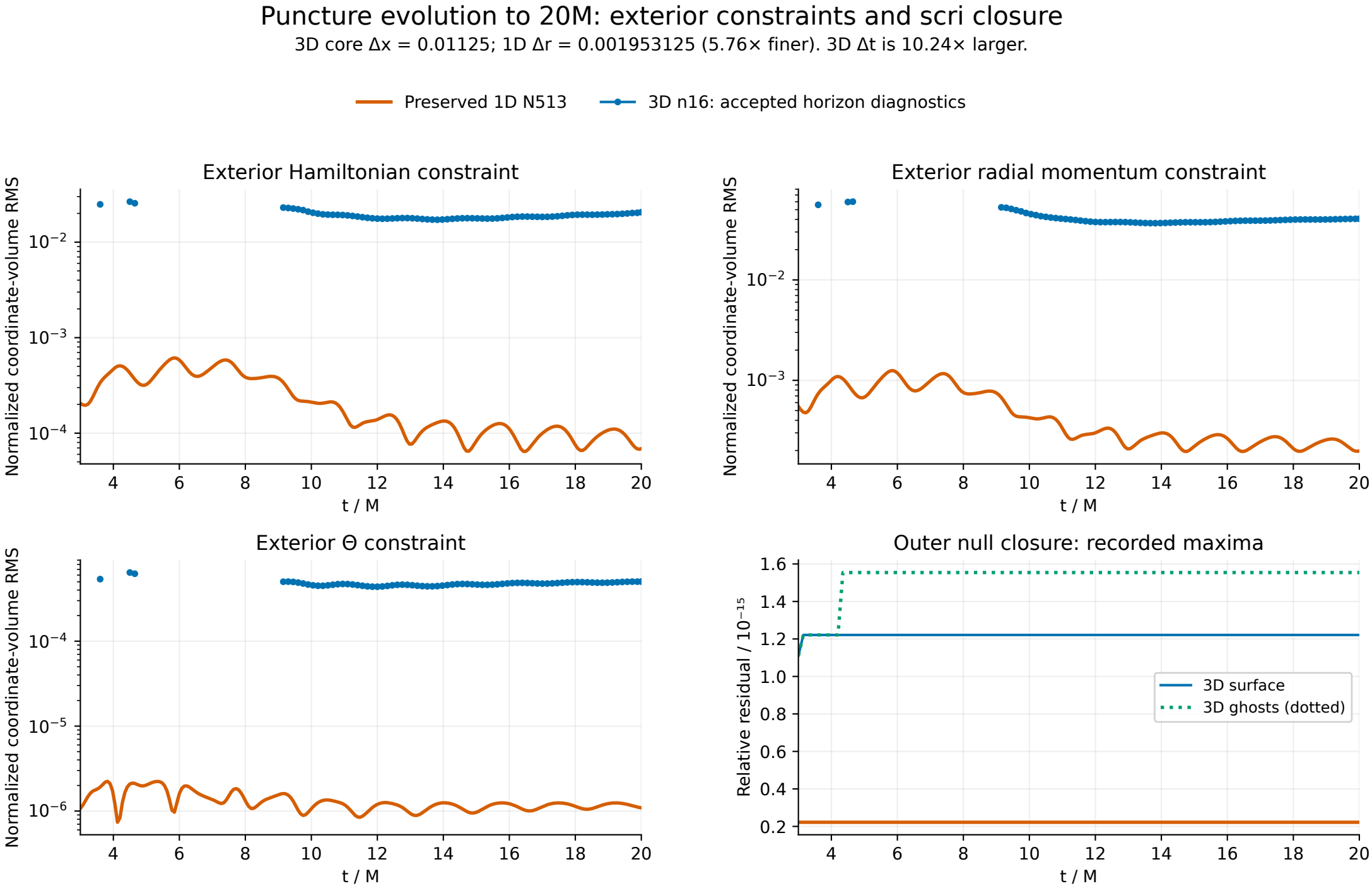
Puncture evolution to 20M: exterior constraints and scri closure
3D core Δx = 0.01125; 1D Δr = 0.001953125 (5.76× finer). 3D Δt is 10.24× larger.
Preserved 1D N513
3D n16: accepted horizon diagnostics
Exterior Hamiltonian constraint
Exterior radial momentum constraint
Exterior Θ constraint
Outer null closure: recorded maxima
Normalized coordinate-volume RMS
Relative residual / 10⁻¹⁵
t / M
3D surface
3D ghosts (dotted)
Each run uses its own horizon cutoff and coordinate-volume quadrature. Missing 3D norms are not interpolated.
3D norms are unavailable at 38 / 114 samples where the horizon tolerance was missed; all samples after 9M are accepted.